

KSEAB PRACTICE QUESTION PAPERS -2025-26

PUC II CHEMISTRY

SIMPLIFIED CHEMISTRY FOR SUCCESS

CHAPTER WISE MLL 2025-2026

**MINIMUM LEVEL LEARNING QUESTIONS (MLL) FOR
PREPARATORY & ANNUAL EXAM-2025-26**

By

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2025-26 BOARD PRACTICE QUESTION PAPERS

Unit -1 SOLUTIONS(12M)

MQP 25-26	PART-A		PART-B	PART-C		PART-D	PART-E	TOTAL MARKS
	MCQ	FIB	2M	INORG	PHY	ORG	PROBLEMS	
MQP-4	01,13	17	-	-	33	-	41,44	12
MQP-5	01,02	16	-	-	32	-	41,43	12
MQP-6	01	-	21	-	34	-	41,46	12

1. The concentration of fluoride ions to be present in water to prevent tooth decay is (P1)

- (a) 1 ppm (b) 1.5 ppm (c) 2 ppm (d) 2.5 ppm

2. Colligative properties depend on _____ (P1)

- (a) the nature of the solute particles dissolved in solution
(b) the number of solute particles in solution
 (c) the physical properties of the solute particles dissolved in solution
 (d) the nature of solvent particles

3. When mercuric iodide is added to the aqueous solutions of potassium iodide then (P2)

- (a) **freezing point is raised.** (b) freezing point is lowered.
 (c) freezing point does not change. (d) boiling point does not change.

4. The concentration of pollutants in water or atmosphere is expressed in terms of (P2)

- (a) μgmL^{-1} (b) molarity (c) molality (d) volume percentage (v/v)

5. The mixture that forms maximum boiling azeotrope at specific concentration is (P3)

- (a) heptane + octane **(b) water + Nitric acid**
 (c) ethanol + water (d) acetone + carbon disulphide

6. The lowering of vapour pressure of a solution causes a lowering of the Freezing Point compared to that of the pure Solvent (P1)

7. The utility or importance of solutions in life depends on their Composition (P2)

8. What is reverse osmosis? Mention one application of it. 2M (P3)

Reverse osmosis: "The process of reversing the direction of osmosis by applying the pressure higher than the osmotic pressure to the solution of higher concentration" is called reverse osmosis.

Application of reverse osmosis: Desalination of sea water.

9. What is semi permeable membrane? Give an example each for natural and synthetic semipermeable membrane. 3M (P1)

Ans: The membranes through which solvent molecules like water can pass but the passage of bigger molecules like solute is hindered are called semipermeable membranes.

Natural semipermeable membrane: Pig's bladder or parchment

Synthetic semipermeable membrane: cellophane

10. State Henry's law. Write its mathematical form. Mention one of its applications.

3M (P2)

Henry's law:

"The partial pressure of gas in vapour phase is directly proportional to the mole fraction of gas in the solution".

i.e., $p \propto x$

$p \rightarrow$ partial pressure of the gas.

$p = K_H \times x$

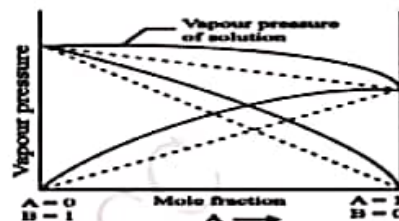
$x \rightarrow$ solubility in mole fraction.

$K_H \rightarrow$ Henry's constant.

"Higher the value of K_H lower the solubility of gas".

11. For this graph; answer the following questions 3M (P3)

- What type of non-ideal solution shows such a behavior?
- Give example for such type of non-ideal solution.
- What type of azeotrope will the mixture of A and B form?



- Ans: a) Non ideal solution with positive deviation.
 b) Acetone and Ethanol
 c) Minimum boiling point azeotrope.

12. The vapour pressure of water is 12.3 k Pa at 300K, Calculate the vapour pressure of 1 molal solution of a non-volatile solute in it.

3M (P1)

Ans: According to the Raoult's law

$$\frac{p_A^0 - p_A}{p_A} = x_B$$

Here p_A = Vapour pressure of solution and p_A^0 = Vapour pressure of solvent

$$\frac{p_A^0 - p_A}{p_A^0} = \frac{n_B}{n_B + n_A} \quad \left(\begin{array}{l} A = H_2O \\ B = \text{solute} \end{array} \right)$$

1 molal means 1 mole of non volatile solute present in 1000 g H_2O

Molar mass of $H_2O = 18 \text{ g mol}^{-1}$

Number of moles in 1000g $H_2O = \frac{1000}{18} = 55.56$ moles.

Number of moles of solute = 1 mol.

$$\therefore \frac{12.3 - p_A}{12.3} = \frac{1}{1 + 55.56}$$

$$12.3 - p_A = 0.0177 \times 12.3$$

$$p_A = 12.3 - 0.2177 = 12.08 \text{ kPa}$$

13. Calculate the molality of 2.5g of ethanoic acid (CH_3COOH) in 75g of benzene. 3M (P1)

Ans: Molality = $\frac{w_B \times 1000}{M_B \times w_A}$

$$w_B = 2.5 \text{ g}, w_A = 75 \text{ g}$$

Molar mass of acetic acid,

$$M_B = 2 \times 12 + 4 \times 1 + 2 \times 16 = 60 \text{ g mol}^{-1}$$

$$\text{Molality} = \frac{2.5 \times 1000}{60 \times 75}$$

$$= 0.5556 \text{ m.}$$

14. The boiling point of benzene is 353.23 K. When 1.80 g of a non-volatile solute was dissolved in 90 g of benzene, the boiling point is raised to 354.11 K. Calculate the molar mass of the solute.

(K_b for the benzene is $2.53 \text{ K kg mol}^{-1}$)

3M (P2)

Ans: $M_B = \frac{K_b \times w_B \times 1000}{\Delta T_b \times w_A}$ But $\Delta T_b = T_b - T_b^*$

$$\therefore M_B = \frac{K_b \times w_B \times 1000}{(T_b - T_b^*) \times w_A}$$

$$M_B = \frac{2.53 \times 1.8 \times 1000}{(354.11 - 353.23) \times 90}$$

$$= \frac{2.53 \times 1.8 \times 1000}{0.88 \times 90} = 57.5 \text{ g mol}^{-1}.$$

15. Calculate the mass of non-volatile solute (molar mass 40 g mol^{-1}) which should be dissolved

in 114 g octane to reduce its vapour pressure 80%. (Molar mass of octane = 114 g mol^{-1}). 3M(P2)

Ans: Molecular mass of octane (C_8H_{18})
 $= M_A = 8 \times 12 + 18 \times 1 = 114 \text{ g mol}^{-1}$

Vapour pressure of octane after dissolving solute $= \frac{80p_A^*}{100} = 0.8p_A^*$

$$\frac{p_A^* - p_A}{p_A^*} = \frac{n_B}{n_A + n_B}$$

$$n_B = \frac{w_B}{40} \text{ mol},$$

$$n_A = \frac{114}{114} = 1 \text{ mol}$$

$$\therefore \frac{p_A^* - 0.8p_A^*}{p_A^*} = \frac{\frac{w_B}{40}}{\left(\frac{w_B}{40} + 1\right)}$$

$$\frac{p_A^* - 0.8p_A^*}{p_A^*} = \frac{w_B}{w_B + 40}$$

$$0.2 = \frac{w_B}{w_B + 40}$$

$$\therefore w_B = 10 \text{ g}$$

16. 100 g of liquid 'A' (molar mass 140 g mol^{-1}) was dissolved in 1000 g of liquid 'B' (molar mass 180 g mol^{-1}). The vapour pressure of pure liquid 'B' was found to be 500 torr. Calculate the vapour pressure of liquid 'A' and its vapour pressure in the solution if the total vapour pressure of the solution is 475 torr.

3M(P3)

Ans: Number of moles of liquid

$$A = n_A = \frac{100}{140} = 0.714 \text{ mol}$$

Number of moles of liquid

$$B = n_B = \frac{1000}{180} = 5.556 \text{ mol}$$

$$x_B = \frac{n_B}{n_A + n_B}$$

$$x_B = \left(\frac{5.556}{5.556 + 0.714} \right) = 0.886$$

$$P_B = P_B^0 x_B = 500 \times 0.886 = 443 \text{ torr}$$

$$P_{\text{total}} = P_A + P_B$$

$$P_A = P_{\text{total}} - P_B = 475 - 443 = 32 \text{ torr}$$

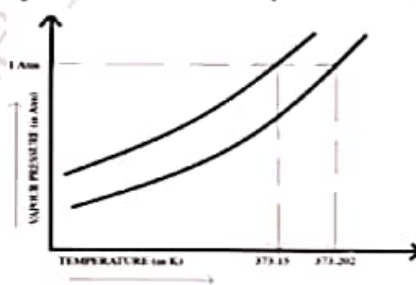
$$x_A = 1 - x_B = 1 - 0.886 = 0.114$$

$$P_A = P_A^0 x_A$$

$$\therefore P_A^0 = \frac{P_A}{x_A}$$

$$= \frac{32}{0.114} = 280.7 \text{ torr}$$

17. The vapour pressure (in Atm) curve for solution containing non-volatile solid substance 'G' and pure solvent is plotted against temperature (in K) is as shown in figure. Calculate the molality of the solution. (K_b for water is $0.52 \text{ K Kg mol}^{-1}$).



Ans: $\Delta T_b = K_b m$

$$m = \frac{\Delta T_b}{K_b}$$

$$= \frac{T_b - T_b^0}{K_b}$$

$$m = \frac{373.202 - 373.15}{0.52} = \frac{0.052}{0.52} = 0.1 \text{ m}$$

2025-26 BOARD PRACTICE QUESTION PAPERS

Unit -2 ELECTROCHEMISTRY(13M)

MQP 25-26	PART-A		PART-B	PART-C		PART-D	PART-E	TOTAL MARKS
	MCQ	FIB	2M	INORG	PHY	ORG	PROBLEMS	
MQP-4	05	-	-	-	32,34	-	43,45	13
MQP-5	03	-	-	-	31,33	-	42,45	13
MQP-6	03	-	-	-	32,33	-	42,43	13

1. **Statement I: Conductivity of all electrolytes decreases on dilution.**

Statement II: On dilution, number of ions per unit volume decreases.

(P1)

- a) Both Statement I and Statement II are correct.
 b) Statement I is incorrect but Statement II is correct.
 c) Both Statement I and Statement II are incorrect.
 d) Statement I is correct but Statement II is incorrect.
2. **The lowest standard reduction potential of lithium electrode for the reduction reaction Li^+ indicates that** **(P2)**
- (a) Lithium is the most powerful oxidising agent
 (b) **Lithium is the most powerful reducing agent**
 (c) Lithium ion is the strongest oxidising agent
 (d) Lithium is the weakest reducing agent

3. **Statement I : Cu displaces H_2 gas from dilute acids.**

Statement II : Cu^{2+} ions get reduced more easily than H^+ ions.

(P3)

- (a) Both Statement I and Statement II are correct.
 (b) Statement I is incorrect but Statement II is correct.
 (c) Both Statement I and Statement II are incorrect.
 (d) Statement I is correct but Statement II is incorrect.

4. **For the standard hydrogen electrode (SHE):**

3M (P1)

i. Write the electrode representation.

ii. What is the concentration of hydrogen ion?

iii. Name the metal used in electrode.

Ans: (i) $\text{Pt(s)} | \text{H}_2(\text{g}) (1 \text{ bar}) | \text{H}^+(\text{aq}) (1\text{M})$

(ii) 1M

(iii) Pt

5. **Mention any three factors affecting electrolytic conductance.**

3M (P1)

Factors affecting the electrolytic or ionic conductance:

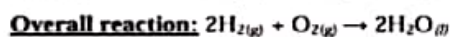
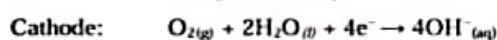
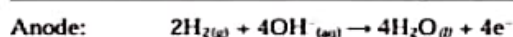
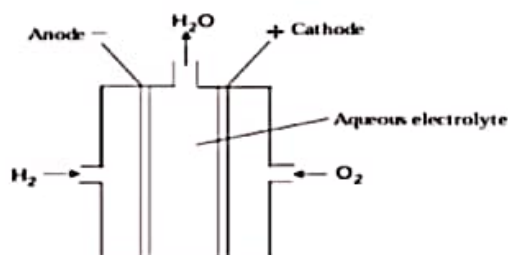
- (i) The nature of the electrolyte
 (ii) The concentration of the electrolyte
 (iii) Temperature

6. Write four differences between metallic conductors and electrolytic conductors. 3M (P2)

Metallic/Electronic Conductors	Ionic/ Electrolytic conductors
• Conduct electricity due to movement of electrons.	• Conduct electricity due to movement of ions.
• Conduction takes place in solid state.	• Conduction takes place in molten state/ in aqueous solutions.
• Metallic conductance decreases with increase in temperature.	• Ionic Conductance increases with increases in temperature.

7. What are fuel cells? Write the anode and cathode reactions of H₂-O₂ fuel cell. 3M (P2)Fuel cells:

Fuel cells are galvanic cells which generate the electrical energy by the combustion of fuels like methane, hydrogen, methanol etc.

Reactions at anode and cathode of H₂ – O₂ fuel cell:Diagram of H₂ – O₂ fuel cell:

8. Draw the neat labelled diagram of Standard Hydrogen Electrode. Write its value of electrode potential and half-cell reaction. 3M (P3)

Standard hydrogen electrode (SHE) or

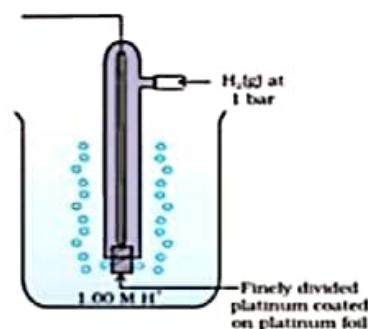
➤ Standard electro potential (E^0) of H₂ electrode 'Zero'

➤ Electrode reaction $\rightarrow \text{H}^{+}_{(aq)} + \text{e}^{-} \rightleftharpoons \frac{1}{2} \text{H}_{2(g)}$

➤ Symbolically representation of SHE is $\text{Pt(s)} / \text{H}_{2(g)} / \text{H}^{+}(\text{aq}, 1\text{M})$

➤ The pressure of hydrogen gas in SHE is 1 bar.

➤ The concentration of hydrogen ion in the solution is 1 molar (1M).



9. State Kohlrausch law. Mention two applications of it. 3M (P3)

Kohlrausch's Law of independent migration of ions:

"Limiting molar conductivity of an electrolyte is the sum of the individual contributions of the cation and anion of the electrolyte."

$$\Lambda^0_m(\text{NaCl}) = \lambda^0_{\text{Na}^+} + \lambda^0_{\text{Cl}^-}$$

Applications of Kohlrausch's Law:

1) Calculation of molar conductivity at infinite dilution (Λ^0_m) for weak electrolyte from experimentally determined values of Λ^0_m of suitable strong electrolytes.

2) Determination of degree of dissociation and dissociation constant of weak electrolyte.

10. The conductivity of 0.00241 M acetic acid is $7.896 \times 10^{-5} \text{ S cm}^{-1}$. Calculate its molar conductivity and if limiting molar conductivity for acetic acid is $390.5 \text{ S cm}^2 \text{ mol}^{-1}$, what is its dissociation constant? 3M (P1)

Ans: Molar conductivity, $\Lambda_m = \frac{\kappa \times 1000}{M}$

$$= \frac{7.896 \times 10^{-5} \text{ S cm}^{-1}}{0.00241 \text{ mol L}^{-1}} \times \frac{1000 \text{ cm}^3}{\text{L}}$$

$$= 32.76 \text{ S cm}^2 \text{ mol}^{-1}$$

Now, $\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$

$$= \frac{32.76 \text{ S cm}^2 \text{ mol}^{-1}}{390.5 \text{ S cm}^2 \text{ mol}^{-1}} = 0.084$$

$\therefore$ Dissociation constant, $K_a = \frac{c\alpha^2}{(1-\alpha)}$

$$= \frac{(0.00241 \text{ mol L}^{-1})(0.084)^2}{(1-0.084)} = 1.86 \times 10^{-4} \text{ mol L}^{-1}$$

11. Calculate the emf of the following cell at 298 K $\text{Sn(s)} | \text{Sn}^{2+} (0.050 \text{ M}) || \text{H}^+ (0.020 \text{ M}) | \text{H}_2(\text{g}) (1 \text{ bar}) | \text{Pt(s)}$;

Given: $E_{\text{Sn}^{2+}/\text{Sn}}^\circ = -0.14 \text{ V}$.

3M(P1)

$n = 2$

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.0591}{n} \log \frac{[\text{Sn}^{2+}]}{[\text{H}^+]^2}$$

$$= 0.14 - \frac{0.0591}{2} \log \frac{0.05}{(0.02)^2}$$

$$= 0.14 - \frac{0.0591}{2} \log 125$$

$$= 0.14 - \frac{0.0591}{2} \times 2.0969$$

$$= 0.14 - 0.062$$

$$= 0.078 \text{ V}$$

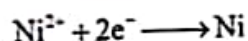
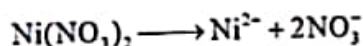
12. A solution of $\text{Ni}(\text{NO}_3)_2$ is electrolysed between platinum electrodes using a current of 10 amperes for 30 minutes. What mass of Ni is deposited at cathode?

(Atomic mass of Ni = 58.7, 1 F = 96500 C)

3M(P2)

Ans: $Q = It$

$$= 10 \times 30 \times 60 = 18000 \text{ C}$$



$$2\text{F} \quad 1 \text{ mole}$$

$$2 \times 96500 \quad 58.7 \text{ g}$$

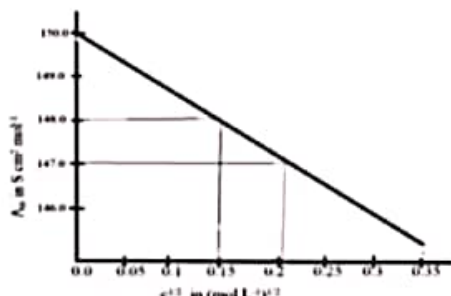
$$2 \times 96500 \text{ deposits} \longrightarrow 58.7 \text{ g Ni}$$

$$18000 \text{ C deposits} \quad x \text{ g Ni}$$

$$\therefore x = \frac{58.7 \times 18000}{2 \times 96500}$$

$$x = 5.4746 \text{ g}$$

13. For strong electrolyte, molar conductivity increases slowly with dilution and represented by equation: $\Lambda_m = \Lambda_m^0 - A c^{1/2}$ The plot of molar conductivity ($\text{S cm}^2 \text{mol}^{-1}$) of strong electrolyte 'X' v/s $c^{1/2} (\text{mol L}^{-1})^{1/2}$ is as shown in the figure. Determine the value of 'A' and limiting molar conductivity for solute 'X' 3M(P2)



Ans: $\Lambda_m = \Lambda_m^0 + (-A)c^{1/2}$

$$y = C + mx$$

$$C = \Lambda_m^0 = \text{Intercept} = 150 \text{ S cm}^2 \text{mol}^{-1} \text{ (from the graph)}$$

$$\text{Slope } A = \frac{\Lambda_2 - \Lambda_1}{c_2^{1/2} - c_1^{1/2}}$$

$$= \frac{148 - 147}{0.24 - 0.15} \text{ (from the graph)}$$

$$= \frac{1 \text{ S cm}^2 \text{mol}^{-1}}{0.09 (\text{mol L}^{-1})^{1/2}} = 11.11 \text{ S cm}^2 \text{mol}^{-1} / (\text{mol L}^{-1})^{1/2}$$

14. Calculate the equilibrium constant of the reaction at 298 K. $E^\circ_{\text{Cell}} = 0.46 \text{ V}$ 3M(P3)



Ans: $E^\circ_{\text{cell}} = \frac{0.059}{n} \log K_c$

$$0.46 = \frac{0.059}{2} \log K_c$$

$$\log K_c = \frac{2 \times 0.46}{0.059}$$

$$\log K_c = 15.5932$$

$$K_c = \text{Antilog}(15.5932)$$

$$K_c = 3.92 \times 10^{15}$$

15. The molar conductivity of 0.025 mol L^{-1} methanoic acid is $46.1 \text{ S cm}^2 \text{mol}^{-1}$. Calculate its degree of dissociation. Given: $\lambda_{\text{H}^+} = 349.6 \text{ S cm}^2 \text{mol}^{-1}$ and $\lambda_{\text{HCOO}^-} = 54.6 \text{ S cm}^2 \text{mol}^{-1}$. 3M(P3)

Ans: $\Lambda^\circ_{\text{m}(\text{HCOOH})} = \lambda^\circ_{\text{H}^+} + \lambda^\circ_{\text{HCOO}^-}$

$$= 349.6 + 54.6$$

$$= 404.2 \text{ S cm}^2 \text{mol}^{-1}$$

$$\text{Degree of dissociation; } \alpha = \frac{\Lambda_{m(\text{HCOOH})}}{\Lambda_{m(\text{HCOOH})}^0}$$

$$= \frac{46.1}{404.2} = 0.114$$

Dissociation constant is calculated by

$$K_a = \frac{c\alpha^2}{(1-\alpha)}$$

$$= \frac{(0.025 \text{ mol L}^{-1})(0.114)^2}{(1-0.114)}$$

$$= 3.67 \times 10^{-4} \text{ mol L}^{-1}$$

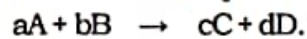
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Unit -3 CHEMICAL KINETICS(12M)

MQP 25-26	PART-A		PART-B	PART-C		PART-D	PART-E	TOTAL MARKS
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MQP-5	04	-	21	-	34	-	44,46	12
MQP-6	02,11	20	-	-	31	-	44,45	12

- The plot of concentration of the reactant vs. time for a reaction is a straight line with a negative slope. The reaction follows a (P1)
 - zero order rate equation
 - first order rate equation
 - second order rate equation
 - third order rate equation
- The rate law for the reaction: $A + B \rightarrow P$ is $\text{Rate} = k[A][B]$. If 'B' is taken large excess, then the order of reaction is (P2)
 - 2
 - 1
 - 0
 - $\frac{1}{2}$
- When the initial concentration of reactant is doubled in a reaction, its half-life period is not affected. Then the order of the reaction is (P3)
 - First
 - Second
 - More than zero but less than first
 - Zero
- In Arrhenius equation $k = Ae^{\frac{-E_a}{RT}}$ the factor $e^{\frac{-E_a}{RT}}$ corresponds to (P3)
 - The fraction of molecules that have kinetic energy greater than energy of Activation.
 - The fraction of molecules that have kinetic energy less than activation energy.
 - The fraction of molecules that have kinetic energy greater than threshold energy.
 - The fraction of molecules that have kinetic energy less than threshold energy.
- Inversion of cane sugar is an Pseudo First order. (P3)
- What are effective collisions? Mention one drawback of collision theory. 2M(P1)
 Ans: The collisions between activated molecules are called effective collisions.
 This theory considers reactants as hard spheres and ignores their structural aspects.
- Define rate of a chemical reaction. How does decrease in concentration of the reactants on the passage of time impact the rate at a given temperature? 2M(P2)
 Ans: Change in concentration of reactant or product in unit time is called rate of a chemical reaction.
 Rate of reaction decreases by decrease in concentration of reactant.

8. State rate law? Write the rate expression and differential rate equation for general reaction. **3M (P1)**



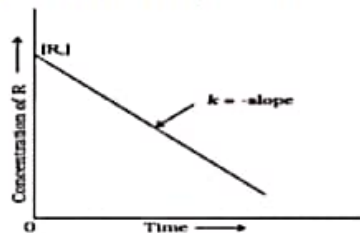
Rate law (law of mass action):

It states that "at constant temperature, rate of a reaction is directly proportional to product of the molar concentration of reactants and each concentration term is raised to suitable powers which may or may not be same as the stoichiometric coefficients of reactants in balanced chemical equation".

$$\Rightarrow -\frac{d[R]}{dt} = k[A]^x[B]^y$$

9. For the following graph, Write the

- Order of reaction,
- Its integrated rate equation.
- SI unit of the rate constant.



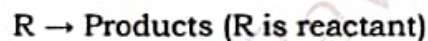
Ans: (i) Zero order reaction

$$(ii) k = \frac{[R]_0 - [R]}{t}$$

$$(iii) \text{mol L}^{-1} \text{s}^{-1}$$

10. Derive an integrated rate equation for the rate constant of a first order reaction. **3M(P3)**

Ans: Consider a first order reaction



According to the rate law,

$$\text{Rate} = k[R]^1$$

$$-\frac{d[R]}{dt} = k[R] \quad \text{'k' is rate constant or velocity constant.}$$

$$\frac{d[R]}{[R]} = -k dt$$

Integrating both sides

$$\int \frac{d[R]}{[R]} = -k \int dt$$

$$\ln [R] = -k t + I \rightarrow (1)$$

At time $t = 0$, $[R] = [R]_0$ $[R]_0$ is the initial concentration of reactant.

$$I = \ln [R]_0 \quad \text{Substitute in equation (1)}$$

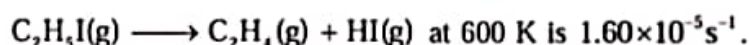
$$\ln [R] = -k t + \ln [R]_0$$

$$k t = \ln [R]_0 - \ln [R]$$

$$k = \frac{1}{t} \ln \frac{[R]_0}{[R]}$$

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

11. The first order rate constant for the decomposition of ethyl iodide by the following reaction



Its energy of activation is 209 kJmol. Calculate the rate constant of the reaction at 700 K.

Ans: We know that $\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_1 \times T_2} \right]$

$$\log \frac{k_2}{1.6 \times 10^{-5}} = \frac{209000 \text{ J mol}^{-1}}{2.303 \times 8.314} \left[\frac{700 - 600}{600 \times 700} \right]$$

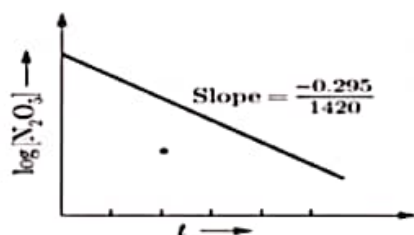
$$\log \frac{k_2}{1.6 \times 10^{-5}} = 2.5989$$

$$\frac{k_2}{1.6 \times 10^{-5}} = \text{Antilog } (2.5989)$$

$$\frac{k_2}{1.6 \times 10^{-5}} = 3.972 \times 10^2$$

$$k_2 = 1.6 \times 10^{-5} \times 3.972 \times 10^2 = 6.3552 \times 10^{-3} \text{ s}^{-1}$$

12. At 318 K, Nitrogen pentoxide decomposes according to the equation; $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$ in gas phase. The plot of $\log [\text{N}_2\text{O}_5]$ v/s time 't' shows a straight line with negative slope as shown in figure. Calculate the rate constant. 3M(P1)



For first order reaction,

$$-\text{slope} = \frac{k}{2.303}$$

$$+\frac{0.295}{1420} = \frac{k}{2.303}$$

$$k = \frac{+0.295 \times 2.303}{1420}$$

$$= 0.000478$$

$$= 4.78 \times 10^{-4} \text{ s}^{-1}$$

13. The rate constants of reactions at 600 K and 700 K are $1.60 \times 10^{-5} \text{ s}^{-1}$ and $6.40 \times 10^{-3} \text{ s}^{-1}$ respectively. Calculate the value of E_a . 3M(P2)

Ans: $\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$

$$\log \frac{6.4 \times 10^{-3}}{1.6 \times 10^{-5}} = \frac{E_a}{2.303 \times 8.314} \left[\frac{700 - 600}{600 \times 700} \right]$$

$$\log 400 = \frac{E_a}{2.303 \times 8.314} \left[\frac{100}{600 \times 700} \right]$$

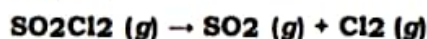
$$2.6021 = \frac{E_a}{2.303 \times 8.314} \times \frac{100}{600 \times 700}$$

$$E_a = \frac{2.6021 \times 2.303 \times 8.314 \times 600 \times 700}{100}$$

$$= 209255.6 \text{ J mol}^{-1}$$

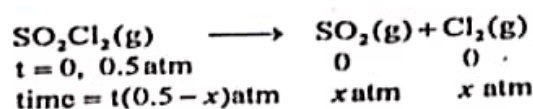
$$= 209.2556 \text{ kJ}$$

14. The following data were obtained the first order thermal decomposition of SO_2Cl_2 at a constant volume: Calculate the rate of the reaction. 3M(P2)



Experiment	Time/s-1	Total pressure/atm
1	0	0.5
2	100	0.6

Ans:



$$p_t = 0.5 - x + x + x$$

$$p_t = 0.5 + x$$

$$x = p_t - 0.5$$

$$p_{\text{SO}_2\text{Cl}_2} = 0.5 - x$$

$$= 0.5 - (p_t - 0.5)$$

$$= 0.5 - p_t + 0.5$$

$$p_{\text{SO}_2\text{Cl}_2} = 1 - p_t$$

$$\text{When } t = 100 \text{ s, } p_t = 0.6$$

$$\therefore p_{\text{SO}_2\text{Cl}_2} = 1 - 0.6 = 0.4 \text{ atm}$$

$$k = \frac{2.303}{t} \log \frac{p_t}{p_\infty}$$

$$= \frac{2.303}{100} \log \frac{0.5}{0.4}$$

$$= \frac{2.303}{100} \log 1.25$$

$$= \frac{2.303}{100} \times 0.0969$$

$$k = 0.00223 \text{ s}^{-1}$$

15. The rate of a reaction quadruples when the temperature changes from 293 K to 313. Calculate the energy of activation of the reaction assuming that it does not change with temperature.

[Given: $R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$]

3M(P3)

Ans: From Arrhenius equation, $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$

It is given that, $k_2 = 4k_1$

$$\log \frac{4k_1}{k_1} = \frac{E_a}{2.303 \times 8.314} \left(\frac{313 - 293}{293 \times 313} \right) (\because k_2 = 4k_1)$$

$$0.6021 = \frac{20 \times E_a}{2.303 \times 8.314 \times 293 \times 313}$$

$$E_a = \frac{0.6021 \times 2.303 \times 8.314 \times 293 \times 313}{20}$$

$$= 52863.33 \text{ J mol}^{-1} = 52.86 \text{ kJ mol}^{-1}$$

16. Sucrose decomposes in acidic solution into glucose and fructose according to the first order

rate law with $t_{1/2} = 3.00$ hrs. What fraction of sample of sucrose remains after 8 hours? 3M(P3)

$$\text{Ans: } k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

The aim is to find $[R]$

$$\text{As } t_{1/2} = 3.0 \text{ hour, } \therefore k = \frac{0.693}{t_{1/2}} = \frac{0.693}{3 \text{ hr}} = 0.231 \text{ hr}^{-1}$$

$$\text{Hence, } 0.231 \text{ hr}^{-1} = \frac{2.303}{8 \text{ hr}} \log \frac{[R]_0}{[R]}$$

$$\log \frac{[R]_0}{[R]} = 0.8024$$

$$\frac{[R]_0}{[R]} = \text{Antilog } (0.8024) = 6.345$$

$$\frac{[R]}{[R]_0} = \frac{1}{6.345} \Rightarrow [R] = \frac{[R]_0}{6.345}$$

$$\therefore [R] = 0.158 \text{ mol L}^{-1}$$

2025-26 BOARD PRACTICE QUESTION PAPERS

Unit -4 d-& f-BLOCK ELEMENTS(10M)

MQP 25-26	PART-A		PART-B	PART-C		PART-D	PART-E	TOTAL MARKS
	MCQ	FIB	2M	INORG	PHY	ORG	PROBLEMS	
MQP-4	02	18		27,30	-	-	-	10
MQP-5	05	17	22	26,27	-	-	-	10
MQP-6	06	18		26,27,28	-	-	-	10

1. In aqueous solution, Electronic configuration of Cu(II) ion is $3d^9$ but electronic configuration of Cu(I) ion is $3d^{10}$, which of the following is correct? (P1)

(a) Cu(II) ion is less stable (b) Cu(II) ion is more stable
(c) Cu(I) and Cu(II) ions are equally stable (d) Cu(I) ion is more stable

2. Lanthanoids react with water to form (P2)

(a) $\text{Ln}(\text{OH})_2$ (b) LnOH (c) Ln_2O_3 (d) $\text{Ln}(\text{OH})_3$

Ans: d) $\text{Ln}(\text{OH})_3$

3. The element having highest magnetic moment is (P3)

(a) Zn (b) Cu (c) Mn (d) Ni

4. Ziegler catalyst is used in manufacture of Polythene. (P1)

5. Spin only magnetic moment of Ti^{2+} is 2.84BM. (P2)

6. Transition metals and their compounds are known for their catalytic activity. Give any two reasons. 2M(P2)

Ans: Transition metals and their compounds used as catalysts because they have,

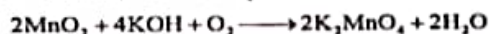
1. Tendency to show variable oxidation states.
2. Tendency to lowering the activation energy by weakening the bonds.
3. Tendency to form reactive intermediates with reactants.

7. A blackish brown coloured solid 'A' when fused with alkali metal hydroxides in presence of air produces a dark green coloured compound 'B' which an electrolytic oxidation in alkaline medium gives a dark purple coloured compound 'C'. Identify A, B and C and write the reactions involved. 3M(P1)

Ans: A = MnO_2

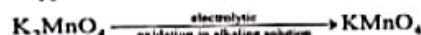
B = K_2MnO_4

C = KMnO_4 or MnO_4^-



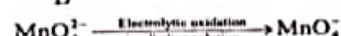
A

B



B

C



B

C

8. In general, transition metal ions of same charge in a given series show progressive decrease in atomic and ionic radius with increase in their atomic number. Give two reasons this change in radius. Name the phenomenon associated with this decrease. 3M (P1)

Ans: Due to the imperfect shielding of one electron by another in the same sub-shell.

It is due to shielding effect of 4f electrons is less than 5d electrons with increase in nuclear charge along the series.

Lanthanide contraction.

9. Among sulphuric acid and hydrochloric acid, which acid is preferred to be used for permanganate titrations? Give reason. 3M (P2)

Ans : Sulphuric acid is preferred because Hydrochloric acid is Oxidised to Chlorine by Permanganate ions.

10. Give reason for the following. 3M (P2)

- Ce(III) is easily oxidised to Ce(IV).
- actinoids show wide range of oxidation states.
- The second and third transition series elements have almost similar atomic radii.

Ans: (i) Because Ce(IV) is favourable by its extra stability of noble gas configuration or to attain f^0 configuration.
(ii) Due to comparable energies between 5f, 6d and 7s levels
(iii) Due to lanthanide contraction.

11. The enthalpies of atomization are given in table below. Select the correct enthalpies of atomization of Zinc and Copper correctly. Justify your answer. 3M (P3)

Enthalpy of atomisation/ kJmol^{-1}
339
130

Ans: Zn = 130

Cu = 339

Enthalpy of atomization of Zn is less. This is because inter-atomic electronic bonding is weak due to the absence of unpaired electrons.

Due to the presence of unpaired electron in Cu, inter atomic electronic bonding is stronger. Hence enthalpy of atomization of Cu is high compare to zinc.

12. (a) Which metal in the first series of transition metal exhibits +1 oxidation state most frequently and why?

(b) What are interstitial compounds?

3M (P3)

Ans: (a) Copper most commonly shows +1 oxidation state because removing the 4s electron gives stable d^{10} configuration.

(b) The compounds in which small atoms like H, C or N are trapped into the (empty space) cavities of the crystal lattices of metals. Such compounds are called interstitial compounds.

13. Write any three uses of lanthanoids and actinoids.

3M (P3)

Ans: 1. Iron and steel are used as construction materials.

2. Some oxides of d-block elements like TiO are used in pigment industry.

3. MnO_2 is used in dry battery cells and Zn and Ni/Cd also required in battery cells.

4. Actinoids are used for nuclear applications like power generation.

2025-26 BOARD PRACTICE QUESTION PAPERS

Unit -5 CO-ORDINATION COMPOUNDS(11M)

MQP 25-26	PART-A		PART-B	PART-C		PART-D	PART-E	TOTAL MARKS
	MCQ	FIB	2M	INORG	PHY	ORG	PROBLEMS	
MQP-4	09,10	-	-	26,28,29	-	-	-	11
MQP-5	06,07	-	-	28,29,30	-	-	-	11
MQP-6	05,12	-	22	29,30	-	-	-	11

1. The stabilisation of coordination compounds due to chelation is called the chelate effect. Which of the following is the most stable complex species? (P1)

- (a) $[\text{Fe}(\text{CO})_5]$ (b) $[\text{Fe}(\text{CN})_6]^{3-}$ (c) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (d) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

Ans: c) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$

2. The counter ion in the coordination compound $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)] \text{Cl}_2$ is (P1)

- (a) Ammine (b) Cobalt (c) Chloride (d) Nitro

3. The arrangement of ligands in the increasing order of ligand field strength based on spectrochemical series is (P2)

- (a) $\text{I}^- < \text{Cl}^- < \text{OH}^- < \text{H}_2\text{O} < \text{CO}$ (b) $\text{CO} < \text{H}_2\text{O} < \text{OH}^- < \text{Cl}^- < \text{I}^-$
 (c) $\text{I}^- < \text{Cl}^- < \text{H}_2\text{O} < \text{OH}^- < \text{CO}$ (d) $\text{H}_2\text{O} < \text{CO} < \text{I}^- < \text{Cl}^- < \text{OH}^-$

4. If ethane-1,2-diamine is progressively added to green coloured $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ in the molar ratios en:Ni, 1:1, 2:1, 3:1, then the correct order of colour changes of complexes would be (P2)

- (a) green, pale blue, blue, colourless (b) green, pale blue, blue, violet
 (c) green, violet, blue, pale blue (d) green, violet, pale blue, blue

5. The correct IUPAC name of $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$ (P3)

- (a) Potassiumtrioxalatochromium(III) (c) Potassiumtrioxalatochromate(II)
 (b) Potassiumtrioxalatochromate(III) (d) Potassiumoxalatochromate (IV)

6. The correct electronic configuration of d^4 coordination entities in a weak ligand is (P3)

- (a) $t_{2g}^3 e_g^1$ (b) $t_{2g}^4 e_g^0$ (c) $t_{2g}^2 e_g^2$ (d) $e_g^3 t_{2g}^1$

7. Chlorophyll is a coordination compound of metal (Magnesium). (P3)

8. What is an ambidentate ligand? Name the type of structural isomerism arises when such ligand presents in the complex. 2M (P3)

Ambidentate ligand: A mono dentate ligand which has two different donor atoms and either of two ligates in the complex.

The co-ordination compounds containing ambidentate ligands shows the 'linkage isomerism'.

Ex: CN^- , NO_2^- , SCN^-

9. Mention the co-ordination number and co-ordination polyhedral of the following complex ions. 3M(P1)

- (i) $[\text{PtCl}_4]^{2-}$ (ii) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ (iii) $[\text{Co}(\text{en})_3]^{3+}$

Ans: (i) 4, square planar
 (ii) 6, Octahedral
 (iii), 6, Octahedral

10. What is the denticity of the ligand in the Fe(III) EDTA complex. Name the atom/atoms of ligand through which it can bind to the central metal ion. 3M(P1)

Ans: Denticity of EDTA in Fe(III) complex is 6 (hexadentate)

EDTA can form six bonds with iron ion from two nitrogen atoms and four oxygen atoms.

11. What are metal carbonyls? How the M-C σ and M-C π bonds are formed in them? 3M(P1)

Ans: Metal carbonyls are the organometallic compounds in which ligand is carbon monoxide (CO).

Bonding in metal carbonyls: In metal carbonyls, the metal- carbon have both σ and π character.

1. M-C σ bond is formed by donation of pair of electrons of carbonyl carbon to the vacant orbital of the metal.
2. M-C π bond is formed by donation of pair of electrons from the filled d orbitals of metal to the vacant antibonding π^* orbital of carbonyl group.
3. The effect of σ bond formation strengthens the π bond and vice- versa. This is called synergic effect.

12. Write any three postulates of Werner's theory of coordination compounds. 3M(P2)

Ans: Postulates (or) Assumptions of Werner theory of coordination compounds:

- 1) In co-ordination compounds, the central metal atom possess two types of valences i.e.,
a) Primary valence b) Secondary valence
- 2) Primary valence is ionisable, variable and corresponds to oxidation state of central metal atom.
- 3) Secondary valence is non-ionisable, non-variable and corresponds to co-ordination number of central metal atom.
- 4) Primary valence satisfied by negative ions, secondary valence satisfied by negative (or) neutral molecules.
- 5) Primary valence is non-directional, secondary valence is directional.

13. With respect to coordination compounds; what are 3M(P2)

(i) isomers (ii) enantiomers (iii) unidentate ligand

Ans: (i) **Isomers:** Two or more complexes having the same chemical formula but differ in their arrangement of ligands are called isomers.

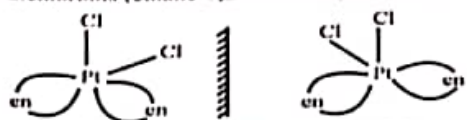
(ii) **Enantiomers:** The isomers which are mirror images that cannot be superimposed on one another are called enantiomers. OR

The isomers which posses non-super imposable mirror image.

(iii) **Unidentate ligand:** A ligand which is bound to a metal ion through a single donar atom is called unidentate ligand

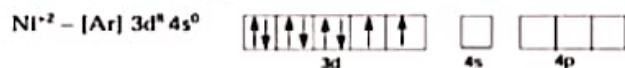
14. Write the IUPAC name of $[\text{PtCl}_2(\text{en})_2]^{2+}$ ion. Draw the optical isomers of it. 3M(P2)

Ans: dichlorobis (ethane-1,2-diamine) platinum(IV)

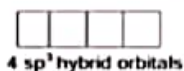
**15. Explain the hybridisation, geometry and magnetic property of $[\text{NiCl}_4]^{2-}$ ion on the basis of Valence Bond Theory. [Atomic mass of Ni = 28] 3M(P3)**

$[\text{NiCl}_4]^{2-}$ [tetrachloridonickelate (II) ion] :

Oxidation state of Ni = +2.



Cl^- is weak ligand, unpaired electrons do not pair up.



Four sp^3 hybrid orbitals filled by electrons donated by 4 Cl^- ligands.



Hybridization - sp^3

Geometry - Tetrahedral

Magnetic nature: Paramagnetic (Due to presence of two unpaired electrons)

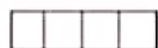
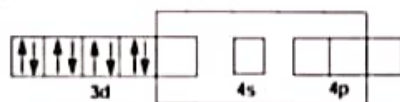
15. Explain the hybridisation, geometry and magnetic property of $[\text{Ni}(\text{CN})_4]^{2-}$ ion on the basis of Valence Bond Theory. [Atomic mass of Ni = 28]

2) $[\text{Ni}(\text{CN})_4]^{2-}$ [tetracyanonickelate (II) ion];

Oxidation state of Ni = +2,



CN^- strong ligand, unpaired electrons get paired.



4 dsp^2 hybrid orbitals

Four dsp^2 hybrid orbitals filled by electrons donated by 4 CN^- ligands.

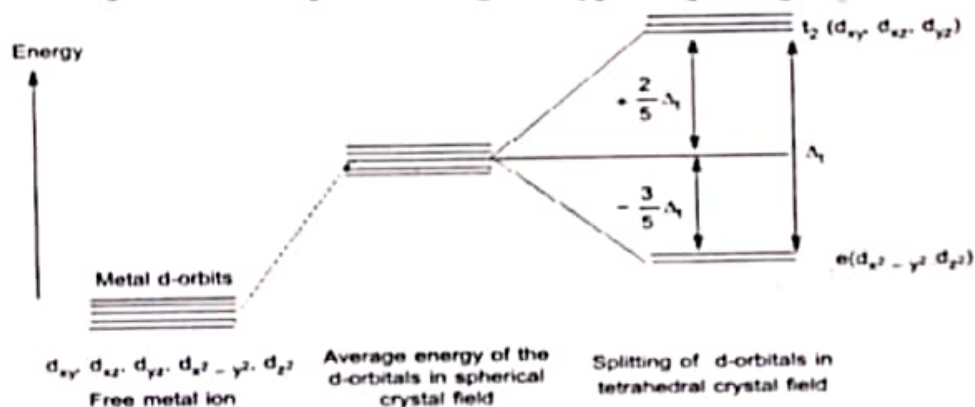


Hybridization- dsp^2

Geometry – Square planar

Magnetic nature: Diamagnetic (Due to absence of unpaired electrons)

16. Draw the energy level diagram for the splitting of d-orbital in tetrahedral crystal field. Give an example metal complex showing this type of splitting. 3M(P3)



Example : $[\text{NiCl}_4]^{2-}$

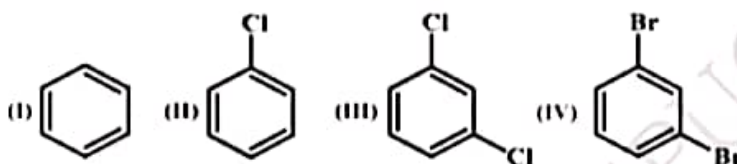
2025-26 BOARD PRACTICE QUESTION PAPERS

Unit -6 HALOALKANES & HALOARENES (09M)

MQP 25-26	PART-A		PART-B	PART-C(3M)		PART-D	PART-E	TOTAL MARKS
	MCQ	FIB	2M	INORG	PHY	ORG(5M)	PROBLEMS	
MQP-4	03	16	22	-	-	36	-	09
MQP-5	11	18	23	-	-	35	-	09
MQP-6	04,13	-	23	-	-	40	-	09

1. Arrange the following compounds in the increasing order of their density

(P1)



- (a) I < III < IV < II
(b) IV < III < II < I
(c) I < II < III < IV
(d) II < IV < III < I
2. The halide that does not undergo nucleophilic substitution with the anion formed by phthalimide is

(P2)

- (a) C_6H_5Cl (b) CH_3Cl (c) $C_6H_5CH_2Cl$ (d) $CH_3CH_2CH_2Cl$

3. Which of the following is an example for vicinal dihalides?

(P3)

- (a) Dichloromethane (b) Allyl chloride (c) Ethylidene dichloride (d) 1,2-Dichloroethane

4. The major product formed when 2-bromopentane reacts with alcoholic KOH

(P3)

- (a) pent-1-ene (b) pent-2-ene (c) pent-2-ol (d) pent-1-ol

5. The chemical used to remove paint is CH_2Cl_2 .

(P1)

6. Freon-12 is manufactured from CCl_4 by Swarts reaction

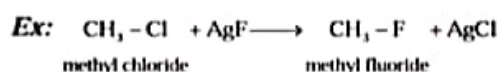
(P2)

7. Explain Swarts reaction with an example.

2M (P1)

Swarts reaction: Alkyl chloride or alkyl bromide on heating with AgF / CoF_3 / Hg_2F_2 /

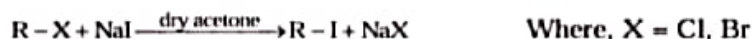
SbF_3 gives alkyl fluoride. This reaction is called Swarts reaction.



8. Explain Finkelstein reaction with an example.

2M (P2)

Finkelstein reaction: Alkyl chloride or alkyl bromide on heating with sodium iodide in dry acetone gives alkyl iodide. This reaction is called Finkelstein's reaction.



9. Toluene on treatment with Cl_2 in sunlight gives benzyl chloride whereas when treated with Cl_2 in dark gives o-chlorobenzene and p-chlorobenzene. Give reason.

2M (P3)

Ans: In sunlight Cl_2 undergoes homolytic cleavage to chlorine radicals $\left(\dot{Cl} \right)$. These radicals abstract a benzylic hydrogen from the methyl group of toluene to give resonance stabilized benzyl radical which combines with $\dot{Cl}$ to give benzyl chloride. OR In presence of sunlight it undergo free radical substitution. In dark, chlorination proceeds by an electrophilic aromatic substitution. The methyl group is in electron releasing activator and directs electrophiles to the ortho and para positions to give o-chlorotoluene and p-chlorotoluene.

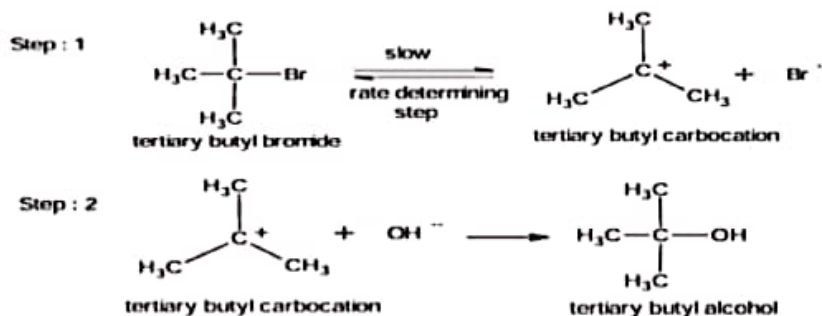
10. (a) Write the equations for the two steps involved in the mechanism for the conversion of 2-bromo-2-methylpropane to 2-methyl-propan-2-ol. Mention the rate determining steps.
 (b) How the plane polarized light is produced? Name the instrument used to measure optical rotation. (3+2) P1

(a) Ans:

S_N¹ reaction (Substitution nucleophilic first order reaction):

Ex: Hydrolysis of tertiary butyl bromide with aqueous NaOH gives tertiary butyl alcohol.

Mechanism:



Rate $\propto$ [alkyl halide], hence S_N¹ reaction is first order reaction.

Rate = k[alkyl halide]

- b) Plane polarized light is produced by passing ordinary light through Nicol prism.
Polarimeter.

11. (a) Give reason for the following:

- 2-chloropropane is an achiral molecule.
- A mixture containing equal proportion of (+)-butan-2-ol and (-)-butan-2-ol shows zero optical rotation.
- 3° alkyl halides undergo S_N1 reactions faster than 2° and 1° alkyl halides

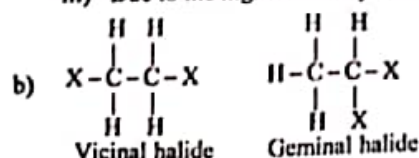
(b) Write the structure of different dihalogen derivatives of Ethane.

(3+2) (P2)

Ans: a) i) Because it does not contain asymmetric carbon atom.

ii) Because optical activity of (+)-butan-2-ol is cancelled by that of (-)-butan-2-ol or due to internal compensations.

iii) Due to the higher stability of 3° carbocations



12. (a) Write the equations for the steps involved in the mechanism for the conversion of tert-butyl bromide to tert-butyl alcohol. Among polar protic solvent and non-polar solvent, name the type of solvent generally used in S_N1 reaction. (3+2) (P3)

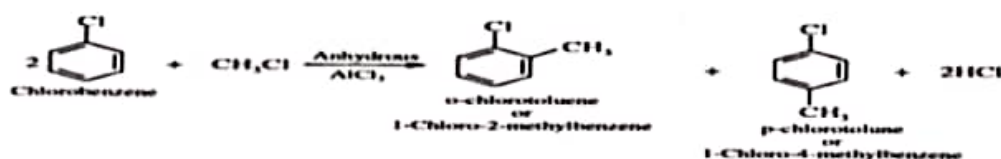
(b) Explain Friedel-Craft's alkylation for chlorobenzene. Give equation.

Ans: (a) Refer above 10th answer S_N1 Mechanism

Polar Protic Solvent like Water, Alcohol, Acetic acid etc.

(b)

Chlorobenzene reacts with alkyl chloride in the presence of anhydrous aluminium chloride to form o-methyl chlorobenzene and p-methyl chlorobenzene.



2025-26 BOARD PRACTICE QUESTION PAPERS

Unit -7 ALCOHOLS,PHENOLS,ETHERS (10M)

MQP 25-26	PART-A		PART-B	PART-C(3M)		PART-D	PART-E	TOTAL MARKS
	MCQ	FIB	2M	INORG	PHY	ORG(5M)	PROBLEMS	
MQP-4	07,08	19	25	-	-	37	-	10
MQP-5	09,10	19	24	-	-	36	-	10
MQP-6	07,09	19	24	-	-	38	-	10

1. The products formed when phenol is treated with dilute nitric acid at low temperature (298K)

is/are

- (i) o-nitrophenol (ii) p-nitrophenol (iii) 2,4,6- trinitrophenol
 (a) only (ii) (b) only (iii)
 (c) all (ii) and (iii) (d) both (i) and (ii)

(P1)

2. Which of the following condition does make the propan-2-ol to undergo dehydration? (P1)

- (a) Conc. H_2SO_4 at 443 K (b) 85% H_3PO_4 at 440 K
 (c) 20% H_3PO_4 at 358 K (d) dil. H_2SO_4 at 400 K

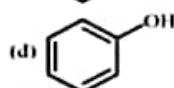
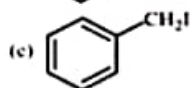
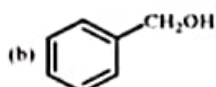
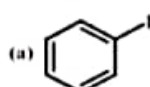
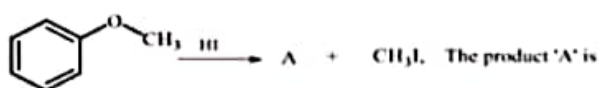
3. Phenol is less acidic than

- (a) ethanol (b) o-nitrophenol (c) o-cresol (d) water

(P2)

4.

(P2)



5. The oxidation of alcohol to aldehydes involves the cleavage of the bond/s

- (a) O - H (b) C - H (c) O - Hand C - H (d) C - C

(P3)

6. Phenol is more acidic than

- (a) Ethanol (b) water (c) o-methyl phenol (d) o-nitrophenol

(P3)

7. CH_3OH is called as Wood alcohol.

(P1)

8. $\text{CH}_2 = \text{CH}-\text{OH}$ is a _____ alcohol.

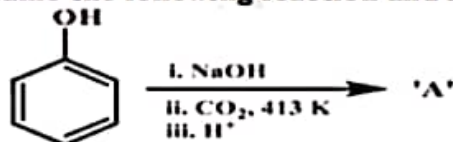
(P2)

9. 2,4,6-trinitrophenol is commonly known as _____

(P3)

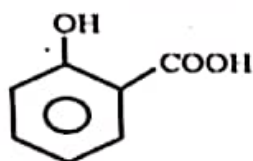
10. Name the following reaction and identify product 'A'.

2M(P1)



Ans:

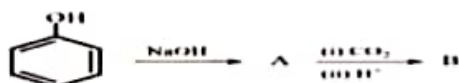
Ans: Kolbe's reaction



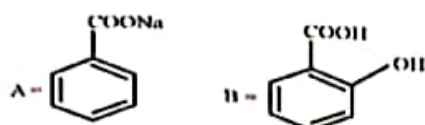
Salicylic acid (2-Hydroxy benzoic acid)

11. Identify 'A' and 'B' in the following reaction

2M(P2)

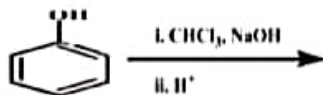


Ans:

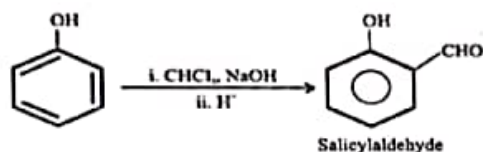


12. Complete the following equation and name the reaction.

2M (P3)



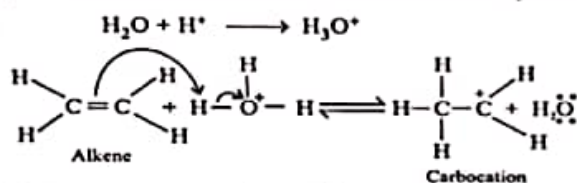
Ans:

13. (a) Write the three equations involved in the mechanism of acid catalyzed hydration of C_2H_4 to C_2H_5OH .

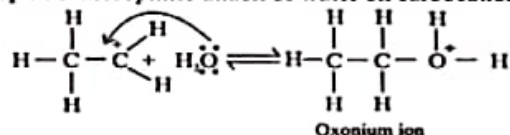
(b) Explain Kolbe's reaction with general equation.

5M(P1)

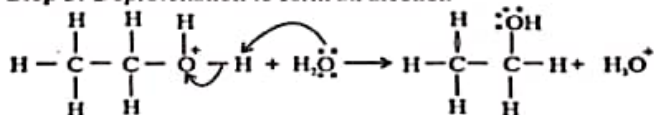
Ans: a) The mechanism of hydration of alkenes involves the following three steps:

Step 1: Protonation of alkene to form carbocation by electrophilic attack of H_3O^+ .

Step 2: Nucleophilic attack of water on carbocation.



Step 3: Deprotonation to form an alcohol.

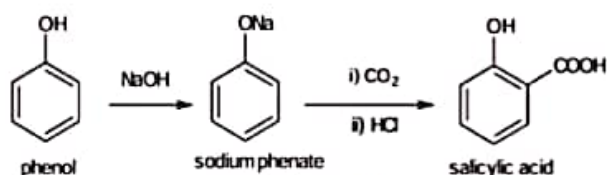


b)

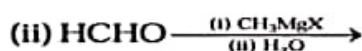
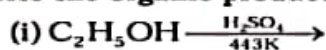
Kolbe's reaction: (Conversion of phenol to salicylic acid):

Sodium phenate heated with carbon dioxide followed by acidification gives salicylic acid.

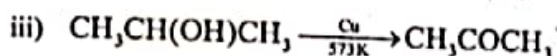
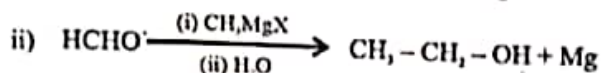
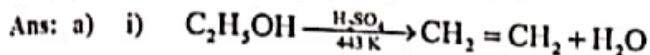
Thos reaction is called Kolbes's reaction.



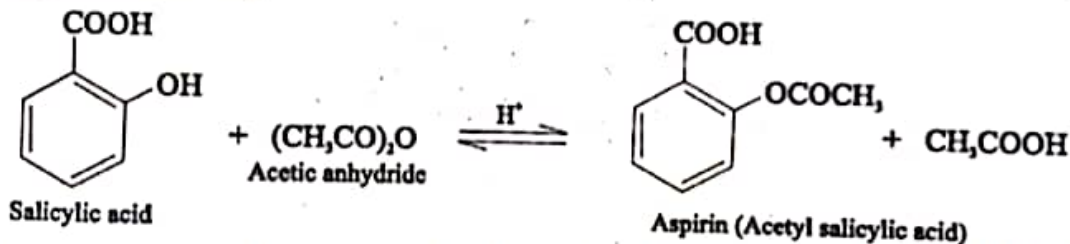
14. (a) Write the organic products of the following reactions.



(b) Write the chemical equation for acetylation of salicylic acid with acetic anhydride. Name the aromatic product of this reaction. 5M (P2)

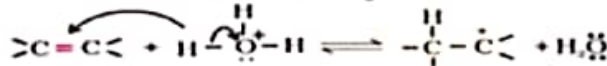


b) Salicylic acid reacts with acetic anhydride in the presence of concentrated sulphuric acid to give aspirin (acetyl salicylic acid).



15. (a) The mechanism of formation of alcohols from alkenes is given below. Identify the errors in the mechanism and rewrite the corrected steps.

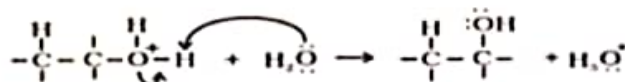
STEP 1



STEP 2



STEP 3



(b) Write the equations for the reaction of concentrated Hydrogen iodide with

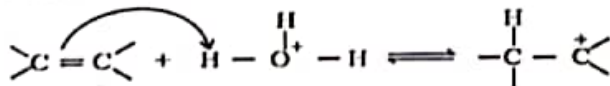
5M (P3)

(i) Methoxybenzene

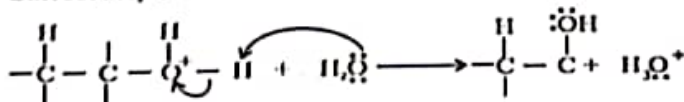
(ii) Methoxyethane

Ans: a) Step 1 and step 3 are wrong

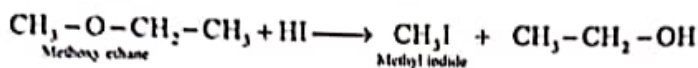
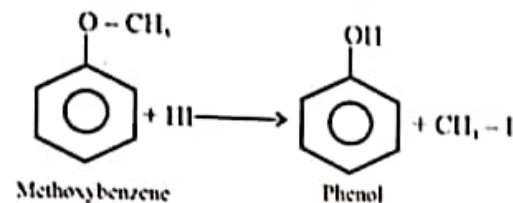
Correct step 1 is



Correct step 3 is



b)



2025-26 BOARD PRACTICE QUESTION PAPERS

Unit -8 ALDEHYDES, KETONES, CARBOXYLIC ACIDS (12M)

MQP 25-26	PART-A		PART-B	PART-C(3M)		PART-D	PART-E	TOTAL MARKS
	MCQ	FIB	2M	INORG	PHY	ORG(5M)	PROBLEMS	
MQP-4	11,12	-	-	-	-	35,40	-	12
MQP-5	08,12	-	-	-	-	37,38	-	12
MQP-6	08,10			-	-	37,39	-	12

1. Match the following given in list-I with List-II.

(P1)

LIST-I	LIST-II
(A) Aldol Condensation	(i) Con NaOH
(B) Cannizaro reaction	(ii) H ₂ / Pd-BaSO ₄
(C) Haloform reaction	(iii) Dilute NaOH
(D) Rosenmund reduction	(iv) NaOI

(a) A-(iv), B-(iii), C-(ii), D-(i)

(b) A-(iii), B-(i), C-(iv), D-(ii)

(c) A-(iii), B-(i), C-(ii), D-(iv)

(d) A-(iii), B-(ii), C-(iv), D-(i)

2. An organic compound with molecular formula C₉H₁₀O₂ gives directly carboxylic acid on hydrolysis and a salt of carboxylic acid on base hydrolysis. The organic compound is

(P1)

(a) 2-ethylbenzoic acid

(b) 2-methyl methyl benzoate

(c) 2-ethylbenzaldehyde

(d) ethyl benzoate

3. The reagent that does not respond both to acetone and benzaldehyde is,

(P2)

(a) sodium hydrogen sulphite

(b) phenyl hydrazine

(c) Fehling's reagent

(d) Grignard reagent

4. The compound that can undergo Hell-Volhard-Zelinsky reaction in the following is

(P2)

(a) CH₃COCH₃

(b) CH₃CHO

(c) (CH₃)₂CHCOOH

(d) C₆H₅COOH

5. What will be the change in hybridization of carbon, when a nucleophile attacks the electrophilic center of the carbonyl group?

(P3)

(a) sp² to sp

(b) sp³ to sp²

(c) sp² to sp³

(d) sp to sp²

6. An organic compound 'X' on treatment with Pyridinium chlorochromate gives compound 'Y'.

Compound 'Y' reacts with I₂ and alkali to form triiodomethane, the compound 'X' is

(P3)

(a) CH₃COCH₃

(b) CH₃CHO

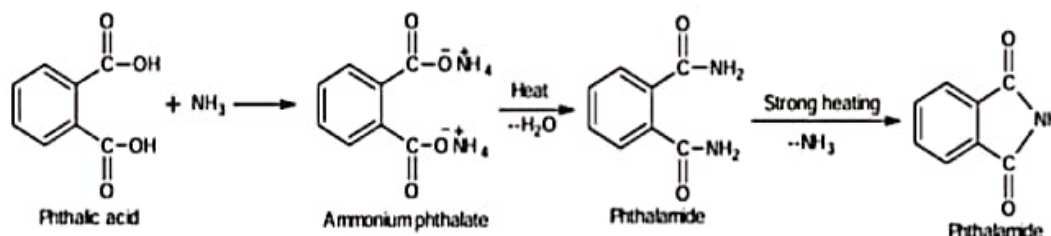
(c) C₂H₅OH

(d) CH₃COOH

7. (a) Write the reactions involved in the conversion of benzene-1,2-dicarboxylic acid to phthalimide.

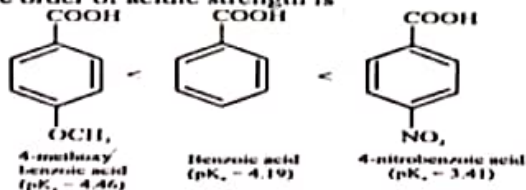
5M(P1)

Conversion of phthalic acid to phthalimide



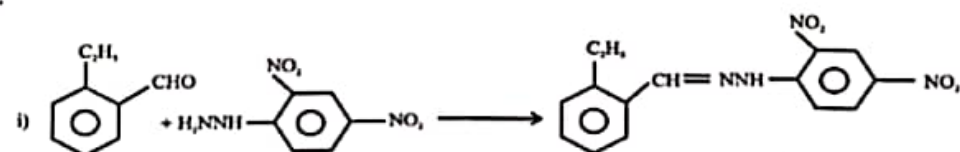
- (b) The pK_a values of 4-methoxybenzoic acid, benzoic acid and 4-nitrobenzoic acid are 4.46, 4.19 and 3.41 respectively. Arrange them in the increasing order of their acid strength. Justify the arrangement.

b) The order of acidic strength is

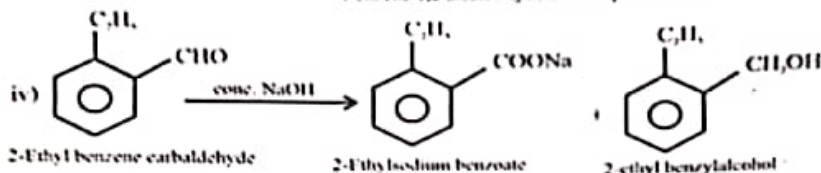
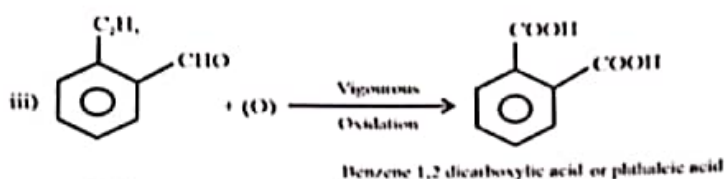
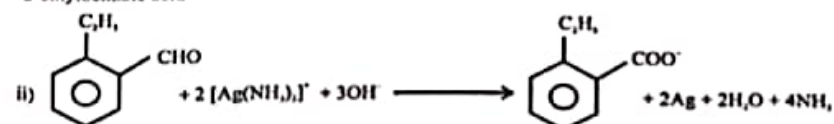


8. An Organic compound with molecular formula $C_9H_{10}O$ forms 2,4-DNP derivative, reduces Tollens reagent and undergoes Cannizzaro reaction. On vigorous oxidation, it gives benzene-1,2-dicarboxylic acid. Identify the compound. Write all the reactions involved. 5M(P1)

Ans:

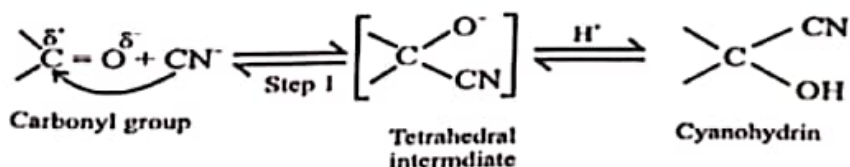
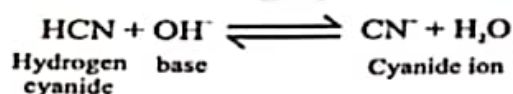


2-ethylbenzoic acid



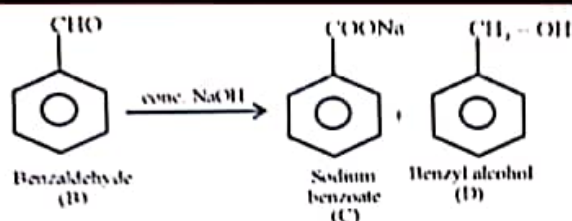
9. (a) Write the steps involved in the mechanism of nucleophilic addition of HCN with aldehyde/ketone.

Ans:

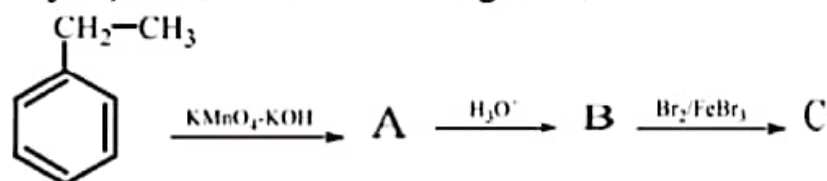


- (b) Compound "A" undergoes Rosenmund reduction to give compound "B" with molecular formula C_7H_6O . compound "B" does not answer Fehling's test. But reacts with conc. NaOH to give compounds "C" and "D". Write two equations involved in this. 5M(P2)

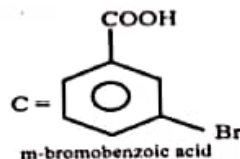
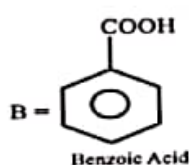
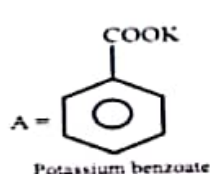




10. (a) Identify 'A', 'B' and 'C' in the following reaction.



Ans: a)



(b) Name any one simple test to distinguish between

5M(P2)

(i) Propanal and Propanone.

(ii) Acetophenone and Benzophenone.

Ans: i) Tollen's Reagent Test

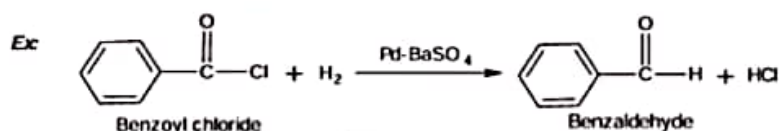
ii) Iodoform Test.

11. (a) How is benzoyl chloride converted to benzaldehyde? Write the chemical equation and name the reaction.

Ans: 5M(P3)

Rosenmund reduction:

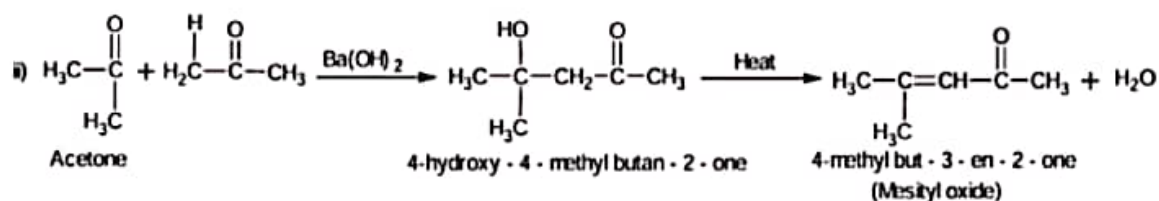
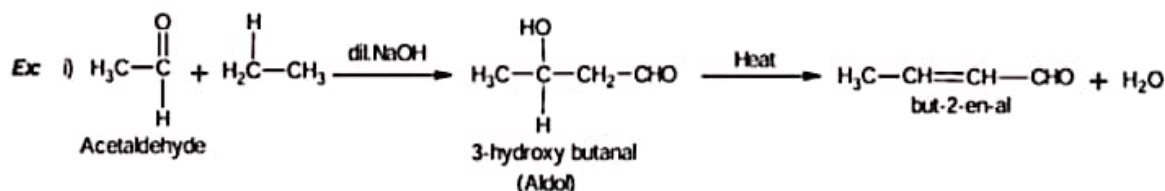
Acid (Acyl) chloride reacts with H_2 in the presence of palladium on barium sulphate gives aldehydes. This reaction is called Rosenmund reduction.



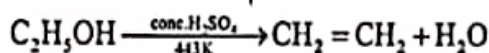
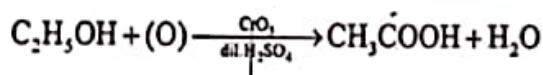
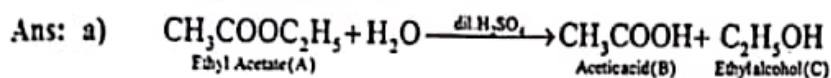
(b) Explain Aldol condensation reaction with an example.

Aldol condensation:

Aldehydes and ketones containing α -hydrogen undergo self-condensation in the presence of dilute alkali to form β -hydroxy aldehyde (aldol) or β -hydroxy ketone (ketol) respectively, which on heating form α, β -unsaturated aldehydes or ketones with elimination of water. This reaction is called aldol condensation.



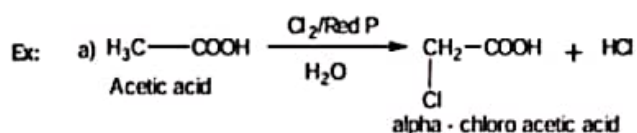
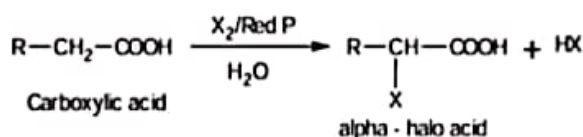
12. (a) An organic compound (A) molecular formula $C_4H_8O_2$ was hydrolysed with dil. sulphuric acid to give a carboxylic acid (B) and an alcohol (C). Oxidation of (C) with Jones reagent produces (B). (C) On dehydration with conc. H_2SO_4 at 443 K gives ethene. Write equations for the reactions involved in this. 5M(P3)



(b) Explain Hell-Volhard-Zelinsky reaction with an example.

Hell-Volhard-Zelinsky reaction or HVZ reaction:

Carboxylic acids with α -hydrogen atom reacts with chlorine or bromine in presence of red phosphorous gives α -halo acids. This reaction is called HVZ reaction.



2025-26 BOARD PRACTICE QUESTION PAPERS

Unit -9 AMINES (07M)

MQP 25-26	PART-A		PART-B	PART-C(3M)		PART-D	PART-E	TOTAL MARKS
	MCQ	FIB	2M	INORG	PHY	ORG(5M)	PROBLEMS	
MQP-4	14,15	-	-	-	-	39	-	07
MQP-5	13,14	-	-	-	-	39	-	07
MQP-6	14	18	-	-	-	36	-	07

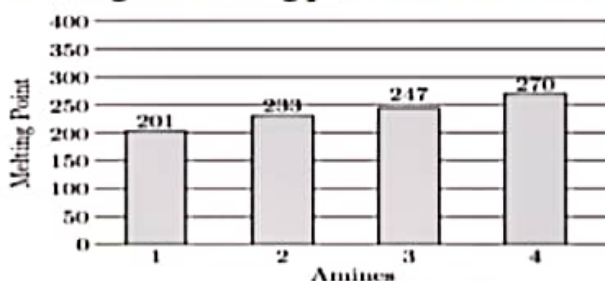
1. An acylation of *N*-ethylethanamine is carried out in the presence of a base stronger than the amine (like pyridine) because it removes

(P1)

- HCl and shifts the equilibrium to the left.
- NaOH and shifts the equilibrium to the right.
- NaOH and shifts the equilibrium to the left.
- HCl and shifts the equilibrium to the right.

2. Study the graph showing the melting points of amines and identify the compounds

(P1)

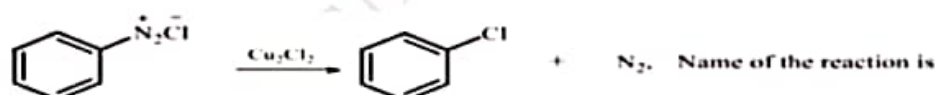


- 1 = Methanamine, 2 = Ethanamine, 3 = Propanamine, 4 = Butanamine
- 1 = Ethanamine, 2 = Propanamine, 3 = Butanamine, 4 = Methanamine
- 1 = Butanamine, 2 = Methanamine, 3 = Propanamine, 4 = Ethanamine
- 1 = Propanamine, 2 = Butanamine, 3 = Methanamine, 4 = Ethanamine

Ans: a.

3.

(P2)



- Swarts reaction
- Wurtz reaction
- Sandmeyer reaction
- Fittig reaction

4. The reagent which is more preferred choice for reducing nitrobenzene to aniline

(P2)

- H_2 / Pd, ethanol
- Fe and HCl
- Sn and HCl
- LiAlH_4

5. Among the following, the most basic compound in aqueous phase is

(P3)

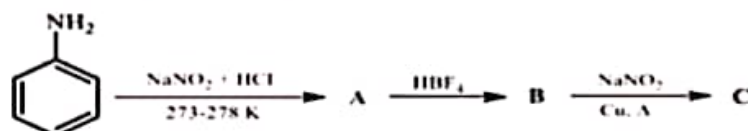
- $(\text{C}_2\text{H}_5)_3\text{NH}$
- $(\text{C}_2\text{H}_5)_3\text{N}$
- $\text{C}_2\text{H}_5\text{NH}_2$
- NH_3

6. *N*-methylmethanamine is an example for

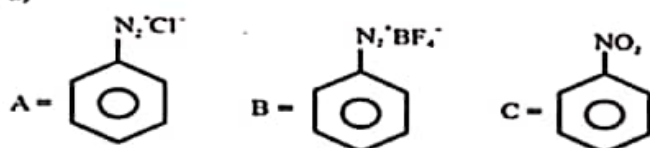
(P3)

7. (a) Identify the products A, B and C in the following.

5M (P1)

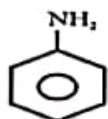


Ans: a)



(b) Mention the IUPAC name of simplest arylamine and write its formula.

b)



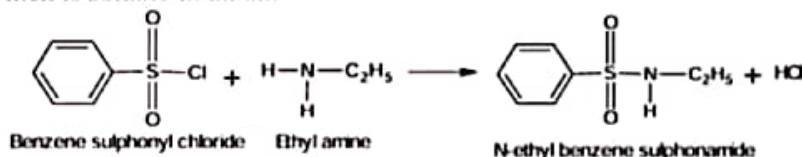
Aniline or Benzenamine

8. (a) What is Hinsberg's reagent? Write the equations for the reaction of primary and secondary amines with Hinsberg's reagent.

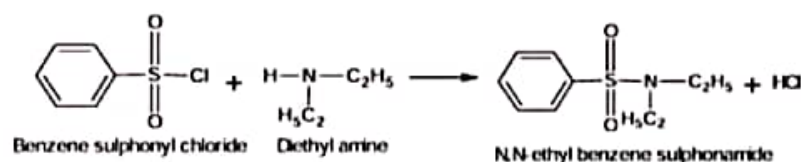
5M(P2)

Ans: Benzene Sulphonyl Chloride ($C_6H_5SO_2Cl$) is called Hinsberg's reagent.

⇒ Primary amine reacts with benzene sulphonyl chloride to form N-ethyl benzene sulphonamide, which is soluble in alkali.



⇒ Secondary amine reacts with benzene sulphonyl chloride to form N, N- diethyl benzene sulphonamide, which is insoluble in alkali.



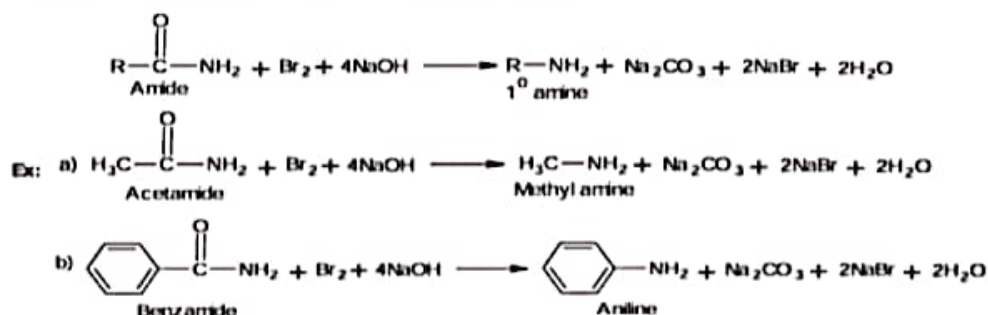
⇒ Tertiary amine does not react with benzene sulphonyl chloride due to absence of H-atom on 'N' atom.

(b) Explain Hoffmann bromamide degradation of benzamide.

Hoffmann bromamide degradation reaction:

Amides on heating with bromine in aqueous or ethanolic solution of NaOH gives 1° amines.

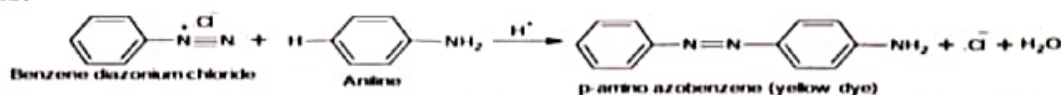
This reaction is called Hoffmann bromamide reaction.



➤ Hoffmann's bromamide reaction is called degradation reaction, because product (amine) obtained is containing one carbon atom less than the amide. So it is called degradation reaction.

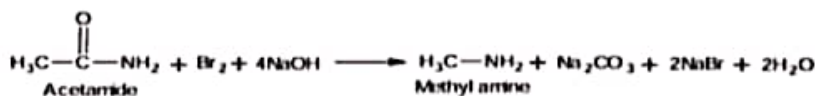
9. (a) How do you convert benzene diazonium chloride to p-aminoazobenzene? Give the equation.

Ans:



(b) How is methanamine prepared by Hoffmann's bromamide degradation? Write equation.

Ans:



(c) Give the IUPAC name of $(CH_3)_2N-C_2H_5$

5M(P3)

Ans: N,N dimethyl ethanamine

2025-26 BOARD PRACTICE QUESTION PAPERS

Unit -10 BIOMOLECULES (09M)

MQP 25-26	PART-A		PART-B	PART-C(3M)		PART-D	PART-E	TOTAL MARKS
	MCQ	FIB	2M	INORG	PHY	ORG(5M)	PROBLEMS	
MQP-4	04	20	24	-	-	38	-	09
MQP-5	15	20	25	-	-	40	-	09
MQP-6	15	16	25	-	-	35	-	09

1. Molecular mass of three amino acids glycine (Gly), Alanine (Ala) and Valine (Val) are 63, 77 and 102 respectively. Molecular mass of tripeptide "Gly-Ala-Val" is (molar mass of water = 18 gmol⁻¹). (P1)
 (a) 206 (b) 242 (c) 260 (d) 188
2. Glucocorticoids are (P2)
 (a) vitamins (b) polypeptide hormones
 (c) amino acid derivatives (d) steroid hormones
3. Match the following given in column I with column II and choose the correct option. (P3)

Column I	Column II
(i) Vitamin A	(A) Rickets
(ii) Vitamin D	(B) Pernicious Anemia
(iii) Vitamin B12	(C) Increased blood clotting time
(iv) Vitamin K	(D) Xerophthalmia

- (a) i-B, ii-D, iii-A, iv-C (b) i-D, ii-A, iii-B, iv-C
 (c) i-D, ii-A, iii-C, iv-B (d) i-A, ii-D, iii-B, iv-C
4. In double helical structure of DNA, the base complementary to adenine is **(Thymine)**. (P1)
5. Glucose reacts with hydroxylamine to form _____. (P2)
6. The hormone which increases sugar level in blood _____. (P3)
7. Name the hormone which is an iodinated derivative of amino acid Tyrosine. How hypothyroidism is controlled? 2M (P1)

Ans : Thyroxine

Hypothyroidism is controlled by adding iodide to commercial salts

8. Name the two components of starch. 2M (P2)

Ans: Amylose and Amylopectine

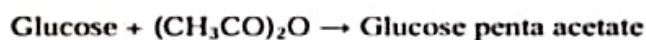
9. Write the chemical equation to show that glucose contains 2M (P3)

(a) Five hydroxyl groups.

(b) Aldehyde group.

Ans: a)

Reaction with acetic anhydride: Glucose reacts with acetic anhydride to give penta-acetyl glucose (glucose penta acetate). This reaction the presence of **five -OH groups** in glucose.



b)

Reaction with bromine water: Glucose undergoes oxidation by mild oxidizing agents like bromine water gives **gluconic acid**. This reaction confirms that carbonyl group in glucose is an **aldehyde**.

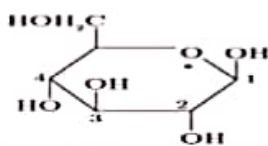


10. (a) Name the two major molecular shapes formed due to the folding of secondary structure of proteins.

Ans: α -helix and β -Pleated sheet Structures

(b) What are the anomers? Find anomeric carbon in given structure.

5M(P1)

**Ans:** The sugar which are differ in their configuration at C-1 carbon atom are called anomers.

Anomeric carbon is C-1 atom.

(c) A person "A" suffering from gums bleeds frequently. The doctor's prescription mentioned that person "A" is suffering from scurvy. Mention any one food source to help person "A" recover faster.

Ans: Food Containing Vitamin C – Citrus fruits, Amla and Green Leafy Vegetables.

11. (a) Answer the following

5M (P2)

- Name the protein present in silk.
- Give an example for an optically inactive alpha amino acid?
- Name the vitamin whose deficiency causes Xerophthalmia.

Ans: i) Keratin

ii) Glycine

iii) Vitamin A

(b) Write the structural and functional difference between DNA and RNA.

Sl. No.	DNA	RNA
i.	It has double stranded helical structure.	It has single stranded polynucleotide chain.
ii.	It controls transmission of genetic information.	It regulates protein synthesis.

12. (a) Identify the carbohydrates and give any one difference between these carbohydrates present in cane sugar and carbohydrate present in milk.

5M (P3)

Ans: a) Cane Sugar – Sucrose - Non Reducing Sugar

Milk – Lactose – Reducing Sugar

(b) What are essential amino acids? Give an example.

Ans: Amino acids that are not synthesized in the body are called essential amino acids.

Ex: Valine, Leucine, Isoleucine, Arginine, Methionine, Phenylalanine, Lysine, Threonine, and Histidine.

(c) Name the nitrogenous base present in RNA but not in DNA.

Ans : Uracil